Mahmoud Hafez

Data and Risk Analyst

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Profile

Data analyst with three years in risk and trading operations at a proprietary trading firm. I analyse trader behaviour, automate reviews and reporting, and build detection tools in Python and SQL. Equally comfortable answering one-off data questions from across the business, from a quick SQL pull to a full breakdown. Lead a back-office and risk team.

Tools

Python	Pandas, NumPy, Matplotlib, Plotly, Streamlit, web scraping with BeautifulSoup
SQL	PostgreSQL, ClickHouse
BI and reporting	Tableau, Power BI, Excel, R and R Markdown
Trading platforms	MetaTrader 5, MatchTrader
Risk analysis	Trading-rule monitoring, martingale, hedging and copy-trading detection, linked-account checks
Automation	Automated reviews, scheduled reports, formatted Excel exports, internal dashboards

Experience

Risk Analyst, Trading Operations

Nov 2023 to present

City Traders Imperium, remote

- Monitor trader behaviour across funded and evaluation accounts against the firm's trading rules.
- Lead the back-office and risk team, track SLAs and productivity, and report to management every week.
- Build the Python and SQL tooling that automates account reviews, detection and reporting (see below).
- Answer one-off data questions from across the business, from quick SQL pulls to full breakdowns.
- Process account upgrades and withdrawals on MT5 and MatchTrader, and escalate technical issues to IT.

First Lieutenant

Sep 2020 to Apr 2023

Egyptian Air Defense Forces

Work at City Traders Imperium

Automated trading-rule review — **90% faster**

Designed and built an automated review covering risk per trade, martingale, hedging, copy trading, one-sided trading, margin usage and the firm's other rules, with a performance scorecard for each account. Cut review time from about 20 minutes to 2.

Group trading detection — **3 minutes**

Designed and built a detector that compares any number of traders, scores how closely each pair is connected and delivers an evidence report in about 3 minutes.

Linked-account detection — **3 minutes**

Designed linked-account detection and co-developed suspicious-connection checks, both now part of the automated account review. A full review averages about 3 minutes.

Weekly management reporting — **about 15 min**

Automated the weekly management report with a Python and SQL pipeline that produces a formatted Excel report in about 15 minutes.

Personal projects

- **Sales analysis, answered in six tools.** A year of sales across 12 CSV files, analysed in Excel, SQL, R, Python, Tableau and Power BI, with recommendations on ad timing, promotion and stock.
- **What drives infant mortality in high-GDP countries.** A Tableau data story comparing infant mortality with health spending and birth rate, 2000 to 2012.
- **Top-rated businesses in the Yelp dataset.** SQL analysis of the top-rated restaurants, shops and spas across several states.

Education

BSc Computer Science, Software Engineering

2015 to 2019

Mansoura University, Egypt

Languages

Arabic (native), English (fluent)

Certificates

- Google Data Analytics, Google
- Python for Everybody, University of Michigan
- SQL for Data Science, UC Davis
- Data Visualization with Tableau, UC Davis
- Python for Data Analysis: Pandas and NumPy, Coursera Project Network
- Getting Started with Power BI Desktop, Coursera Project Network